QUIDEL – Analyst Update (AACC 2016)
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(Transcription by www.RealtimeTranscription.com)

DOUG BRYANT: I said good afternoon. It's still good morning, isn't it?

Well, it's certainly good morning for those of us from San Diego, and 8:00 a.m. is too early for me to have a sandwich. But welcome. We appreciate you being here. Joining me today are Ruben Argueta, our Director of Investor Relations. That's him in the back on his iPhone.

Also here, Mr. Bill Ferenczy, Senior Director and General Manager of our Savanna program, and you'll meet Bill here shortly. He'll be doing a presentation for us.

Also here, Randy Steward, Chief Financial Officer. There he is. In his mind, you'd be applauding right now.

(Laughter)

(Applause)

DOUG BRYANT: Chief Financial Officers don't get enough love, I'm told. But Randy is going to spend some time going over the financials for us, and we'll also talk about M&A and our history, our cash position, etc.

Mr. Mike Abney, Senior Vice President of Distribution sitting in the corner. Mr. Rob Bujarski, Senior Vice President Business Development and General Counsel. He's the guy who hasn't done a transaction recently.

(Laughter)

DOUG BRYANT: It's fine. We're having a fun time, okay? We're not going to ask Rob to present.

(Laughter)

DOUG BRYANT: Mr. Ratan Borkar who is our Vice President of International Commercial Operations all the way there in the back is joining us as well.

So, this is our safe harbor statement. We will be making forward-looking statements. Our strategy, as I think all of you know, has remained unchanged. And today what we hope to do is reframe it slightly within the context of where we are today, and also how we talk about it, what's the vernacular that we use to talk about what we're doing internally.

We say fortify, attack, and forge ahead. I'll talk about what we mean by that, and for a few minutes about Q2. I assume most of you either listened to the call or saw the transcript. We just had our earnings call last week. So I'll just review briefly, and certainly anything you want to ask, additionally, about the quarter we can talk about during Q&A. We'll leave quite a bit of time there.

I actually plan on talking only 20 minutes or so, so that we can make sure that we have enough time at the back end for Q&A.

I'm also going to talk about our business inside influenza and outside influenza. In other words, I want to talk about our non-flu franchises. In the near term, our bigger non-flu growth drivers are, of course, Sofia, Sofia 2, and also Solana. If you were to look at our strategic plan, you would see that within the next couple years, we actually have gated quite a bit of revenue growth in those two product lines.

Following that, though, if you were to look at that same strategic plan in the out years, you would see the uptick of revenue associated with Savanna, which we expect to be the next big driver of non-flu growth.

And as you'll see in Bill's presentation, we expect it to be complementary to Solana. So, while we're certainly not waiting for Savanna before making progress towards building a broader-base diagnostic company, I think that we're near enough with that particular program and set of products that we should spend a bit more time on how Savanna fits with our longer-term strategy, something we really haven't discussed in detail with any audience to this point. So this is the first time we'll unveil what we think we can do relative to what we should say in front of competitors. We will certainly talk about that.

So, we've got Savanna here. We'll be doing a demo if you like during the Q&A. And we have two Sofia 2s in the back, and Ruben will be sharing with you whatever you'd like to know about Sofia 2. And I'll just say that it has to be simple if Ruben's doing the demo. Okay?

(Laughter)

DOUG BRYANT: Given that he's not a scientist, he's not a laboratorian. Okay. He's just our IR guy. And I mentioned that Randy's going to go over the financials as well and talk a little bit about M&A, and then we'll wrap up with Q&A.

So, internally, we describe our strategy as fortify, attack, and forge ahead. Those are good words for organizations internally. I'll spend a couple minutes here in a second just defining what we mean and which programs fit into which of those categories. But first, I'll restate our strategic intent, which is only slightly different than it was in 2009. You know, we've tweaked some words here and there, but essentially the strategy has remained the same.

Our intent is to build a broader-based diagnostics company that delivers revenue and margin more consistently. And we're going to use existing resources to fortify price and volume in our core businesses, while using our capabilities, particularly our molecular capabilities, to attack the limitations of our competitors; and then to forge ahead to create new markets, new products, new ideas, new things that are unique and can actually change the landscape, actually change the way we compete in the space. And we have a couple ideas of things that we're already working on in that regard.

So, there are three concepts, again: Fortify, attack, and forge ahead. Under fortify, obviously, you know, when many of us were on board in 2009, we had a product line called Quick View, a very strong brand, quality product. But it had its limitations, and we thought that we were vulnerable to private label at the time, because of price. But we also thought, given the performance of those products, in particular the sensitivity of those products, that we needed to do something to create a better-performing product, particularly in advance of what we saw as a threat with products that were coming in the molecular space.

So we developed Sofia, and we dramatically improved the performance. In fact, we had a two-log improvement in terms of the sensitivity of the product. And we now have sensitivities in the 90srelative to what some people said were sensitivities in the 80s, and others even said it was worse than that. So we dramatically improved the product, and I think that was the basis for protecting our price and volume. It's also been the basis for us gaining share and also being a participant in growing the market.

We think that having an instrumented system, upgrading from a visually read product to an instrument would be worthwhile, and would cause more and more physicians and other labs to actually do testing in situations where previously they had not.

So, in terms of fortifying, we've done a nice job. And in the process, we've also gained share, as we've talked about many times before.

Attacking competitors. Every competitor and every competitor's product has certain capabilities, and it has certain limitations. That is true absolutely of anything that's in the market today, or ever will be in the market moving forward. And I would say customer needs vary. They vary by individual customer, of course, but they vary by segment, location, and sample type, etc. We've designed products that were designed specifically to address some limitation. So, for example, the Lyra product line, which are reagents that are developed, manufactured, and provided to customers so that they can run those on their own thermo cycling equipment, their own molecular equipment that they have in their larger molecular labs -- these are higher-volume customers that run things in big batches.

Our Lyra product line addresses the problem that these customers have, because these things quite often run on dry ice. They're not stable. You have to create your own master mix, and you have to do all these things in preparation to do a run, and then there's a lot of waste.

So we developed products that were actually lyophilized, they were freeze-dried and stored at room temperature and/or in a regular refrigerator, versus at - 70 or 80° as they are in a molecular lab. So, specifically we designed a product to discuss that limitation that we saw. We've had some success. It hasn't been terrific, because a lot of these customers, frankly, make their own reagents. The same is true with AmpliVue. Customers aren't prepared to acquire instrumentation for their move to molecular methods, so we designed a product called AmpliVue, which is the world's first hand-held disposable molecular device.

And, again, we've done pretty well there. But, again, it's definitely a niche. Most of our customers have it because they run around second or third shift.

And then Solana. Solana, though, I would say, has a much broader appeal. Solana we developed for tests that come in batches, like confirmatory group A strep, or group B strep. Those were both things that are batched. Influenza, today, batched. Right? C. diff, batched. And so there are several others as well where that's true, and we're developing products for Solana that the customer will prefer to run 10, 11, 12 at a time.

And in lab situations where cost really does matter -- there are many lab directors that have a budget within which they have to operate. And coming in with a different platform -- even though it's slightly better in terms of performance -- that blows their lab budget creates a problem for them. They have to go find it somewhere else. What we've been able to do with Solana and group A strep, for example, we can price it at a level that enables a lab director in a micro-lab to move from culture over to a molecular method without increasing their cost dramatically. And so far that's been quite successful for us.

So, forging ahead. We think that it's useful to think about things that would actually change how things are done or change where tests are done. And we've got several ideas that we've been working on.

We think, for example, that with the cloud-based system called Virena that we developed that enables individuals and physicians to see a test result within the context of prevalence, is unique. And all the sudden now the data becomes more important than the individual test result. That's enabled us, by the way, to move more dramatically into the alternate site side of the business, because these customers, more than anything else, want the data.

In one situation, we have a customer that actually tests, or intends to test for flu in the upcoming season. This is in a grocery store chain. And they will charge for the test, and then if it's positive, they'll give the prescription for Tamiflu to the patient for free. So there are unique models coming out because the customer wants the test data so they can use it in a number of different ways, including supply chain.

We're also working on next gen immunoassays that would take the assays to the confirmatory level. We also have a product in Sofia 2 that takes Sofia with a lower cost and smaller footprint, enables us to penetrate a far greater set of professional market than we had anticipated previously.

So, in total, we have a number of things across the three categories. I think Sofia 2, as maybe you'll learn a little bit more during Q&A, actually helps us address each of those particular strategies.

So we just had our earnings call last week, so -- but I did want to say that Q2 in particular is interesting, because it's typically either a non-flu or a low-flu quarter. So for us it's a good snapshot that tells us how our non-flu business is doing at that moment in time. So I won't spend a great deal of time on this particular slide, because it's basically the financial recap. But we did, in the second quarter, grow 11% over the prior year QT2 to 39 million. And 39.1 million is a record for our company in Q2. 11 million of that 39 million was influenza, and 28 million of that was other products other than influenza.

We had 9.3 million in new product revenue, which is basically Sofia and molecular products in the quarter. For the six months, it's not as useful, because we had a soft Q1, to look at the six-month. But including that soft Q1, we did only just under 90 million in revenue. Only 31 million of that was influenza, which is quite low for a six-month period for us, and 24.9 million new product revenue, that was all Sofia and all molecular products. And then trailing 12 same issue, because Q4, Q1 was soft in terms of influenza, we were at about 189 million for the trailing 12. Probably would've been north of 200 with any normal sort of flu season.

We had only 77 million in flu sales the last 12 months, which I'll show you, versus expectations, what we normally think we would do. And then finally, 59.3 in new product revenue. We did have several regulatory submissions and approvals in the quarter, a few of those international, and also here in the U.S.

We did put in our latest high-speed manufacturing line in San Diego. That gives us an additional 20 million test capacity for Sofia cartridge manufacturing, and also puts in place for us a permanent cost reduction in terms of our manufacturing cost.

We did close in the quarter a number of large Solana and large Sofia and Virena customers. We had a thousand or so Virena installs. We think that will accelerate as we exit this year, that we'll end the year somewhere north of 4,000, which would be -- we think rich enough to provide all the things that would power the data visualization tools that we've developed and are launching in a couple key markets right now.

We think the alternate site strategy is finally taking off within -- and I think within a couple years we will see meaningful and large numbers of people seeking treatment in nonprofessional segments. And as they do, we think that that's good for us, and for those of us who are in those small labs in those facilities.

So, as we move forward, tracking progress and the key non-flu franchises will become increasingly important for us as we look to lessen the volatility of our revenues and margins, and hopefully improve our share price multiples over time.

So, I wanted to start that discussion on the non-flu piece by talking about flu, but in a provocative way. If you look at the -- this slide, on the Y axis you have percentage influenza-like illness, just as you would recognize from the CDC reports, right?

And then each -- the red line is for each of the weeks, is that percentage over time. And so you can see that all the way back to 2009, the softness here was followed by the pandemic with the H1N1 California strain. And then a peak again, and then bad season, good season, almost good season, good season, not-so-good season. Right?

Okay. So that's volatility of influenza. True or false? Right? I mean, we can see it. We can chart it out. You can see that we don't always have flu. There's no such thing as normal, at least not normal several years in a row, right?

So, having said all that, then, in the blue dots are the revenue corresponding over here on the right-hand Y axis, the revenue for the year. This is 2010. This is the end of 2010. That's 2010 revenue, right?

So what's different about 2010? 2010 you have 11 months worth of DHI revenue in there, right? So moving forward, I'm able to see what my actual organic growth rate -- except for we did some small acquisitions like BioHelix and others. But for the most part this is organic growth in the midst of volatility of flu.

So, while ILI varies with each season, Quidel's revenues have grown at 12% through growth in influenza, test sales, and the introduction of non-flu products that capture market share. So, the reason that we're able to grow 12%, even though it's extremely volatile, is because we've done so by capturing share. If we hadn't captured share, our revenue would be all over the place, right? But the reason this works for us is that we have captured share, and we plan on capturing share moving forward, at least for several years that we see in front of us.

So, am I suggesting that we not focus growing non-flu revenue, both organically and through acquisition? Is that what this slide is meant to suggest? No, of course not. But it is interesting that with the volatility, we're still growing consistently. But, again, it's because we're growing share. Make sense?

So, volatility is bad, but it's not that bad when you're doing other things to grow your business.

Okay. But, having said all that, Q2 does provide visibility into growth and key non-flu franchises. And so for reference I list flu there in terms of annual revenue. And at this point, given the share that we have, you would expect us in a given year to have 85 to 95 million in flu revenue, okay?

Clearly, we continue to gain share. That number is not appropriate. It's going to be slightly higher. But in this particular quarter, we grew 21% over the prior year quarter. Some of that was market share gain, some of that was the fact that we had low distributor inventories. But frankly, they were pretty much low last year as well. So some market share gain, but a lot of it was spillover from Q1 into Q2 with influenza B. There wasn't any A, but there was some B in a handful of states, and that's what drove distributors to need to reorder in the quarter.

So -- but getting back to non-flu. So the total non-flu, we think, is somewhere between 100-110, because there are some products that are tied in a minor way to a respiratory season. But we grew that business 8%. Immunoassay for strep is a $30 million business now, and it grew 13%. RSV is around a $10 million business for us now. That grew 29%.

Graves' Disease is now a $10 million business. Continuing to grow at 8%. We would expect it would grow around 8% moving forward, until we introduce our next product, which is TBI, a blocking antibody for Hashimoto's Disease. And then the life science research segment -- we list it that way because that's the customer, but this is really our bone health and complement markers plus some renal markers, as well. It includes now also the Immutopics products which came as a result of a recent small acquisition. And we would expect that business normally to grow around the 10% range moving forward. But obviously with the new sales from Immutopics, it's higher, and there you see it at 57% for the quarter.

Our molecular franchise is finally now over $10 million, thanks to Solana. I will suggest to you that what's in the pipeline for Solana is far more significant than were the sales in Q2, as we had a number of customers coming online. But that's gone pretty well for us. And I would say we're doing far better in the physician office segment than I had thought we would.

And then finally, hCG is a consistent business for us and grew modestly about 4% in the quarter. So, you can see that in sum, those are the larger things that are north of $10 million or so that are driving our non-flu organic strategy.

Okay. Here's how we look at near-term non-flu growth drivers. For Sofia and Sofia 2, in the near term, we would expect strep A. We would expect RSV. We'll have a 5-10K version for both Lyme and Vitamin D imminently, followed by a whole blood finger stick CLIA waived product after that.

And I would expect in the near term you'll see evidence that we're moving business into the adjacent markets into pharmacies. Urgent cares are now a huge category, over 7,000 urgent cares. And many of our very large closes have been in that segment. Many grocery stores now have pharmacies, and that's a big category for us now. But, basically, Sofia and Sofia 2 moving forward are and expected to be in lots of different places. You know, we're in places that, you know, they're small numbers, but we're on Navy ships and border patrol and in airports and all sorts of places. But the big drivers, I think, are going to be the urgent care centers and the pharmacies.

The retail clinics, for whatever set of reasons, I don't understand the strategy, have not really grown like we thought that they would, and I'm not sure why that is. But I think some of it has to do with where they are in states and with the collaborative practice agreements actually say that pharmacists can and cannot do.

Bio-assays, I already mentioned TBI. We'll launch that next year. For Solana, as you know, we have group A strep, but we will also have shortly group A plus C or G. Many of the large hospitals in particular, the children's hospitals, want C or G as well. So we'll have that assay.

On the call last week I talked about the submission of trichomonas, and I also talked about HSV 1 and 2/VZV. We also have a group A strep and a C. diff assay in development. We're in the research phase looking at CT/NG.

Those will be online here shortly if you don't want to take a shot, it's up to you. Ruben will put them online so you can have them, the slides.

I mentioned the research segment already. We're also now entering the world of CROs by developing custom cells for their cell culture projects tied to pharmaceutical research. That's just taking off. But I expect that to be a contributor moving forward. And then finally, I'll leave it to Randy to talk about -- Randy and Rob, or Rob, to talk about M&A or the lack of M&A. And I'm just joking with Rob. That's all he does, just looking and looking and looking.

(Laughter)

DOUG BRYANT: So, following the successful launch of Sofia 2 and Solana, which are, as I mentioned before, the key non-flu growth drivers, we will then launch Savanna. And I've asked Bill Ferenczy to talk about Savanna's capabilities and its limitations, and our plans to address the limitations of our molecular competitors, those that are here today and those expected to be with us moving forward. And so I'll just ask Bill to come up and get started. Take your time, Bill, and tell them about Savanna.

BILL FERENCZY: All right. Am I working? Can everybody hear me? Okay.

Thanks, Doug.

I joined Quidel just about five years ago, I think today, might have been actually the day. It's pretty close to my fifth anniversary. I'm doing better at that than my wedding anniversary, which my wife has to remind me. But I joined as the Director of Marketing for the U.S. and was here and got to enjoy the launches of Sofia and AmpliVue and the other new products that Doug has referenced. Just a little under two years ago, though, Doug asked me to join the Savanna program with, really, the primary goal to lead its commercialization.

And during the initial phase, I spent a lot of time trying to understand viral load testing in Africa, the kind of primary mission, and to provide voice of customer and other feedback into the development team so we could finalize the design.

More recently I've shifted gears pretty heavily to looking at what we can do with Savanna in the U.S. and other developed areas. And so what I'm going to do here today is go through a number of things. One, I'm going to give you some background on what I call the African legacy, and that's so you can understand the design of the system and why it's built the way it's built.

And I'm going to walk, then, through that. I'll walk you kind of through a general overview of the system. And then, as Doug has mentioned a couple of times, we focused on thoroughly understanding the capabilities and the limitations so that we can then apply what we know about it to decide where we're going to go and how we're going to launch it. And so I'll talk very briefly about what we're going to do with respect to Africa, and then focus more heavily on the approach we're going to take for the U.S. and really, other developed countries. And then very briefly at the end, just kind of touch upon what we see as the potential market opportunities in the areas that we target.

So, again, Savanna really had its origin as a global health mission starting at Northwestern University out of the engineering school, or the center there for innovation and global health, led by a guy, Dave Kelso, who has a long history of success in diagnostics and kind of retired into the university. And He was getting a lot of insight and direction from the Global Health Foundation led by Kara. We stepped in to be the commercial partner and were provided $20 million in funding from the Gates Foundation in order to develop Savanna for its viral load mission in Africa.

It's really important to understand that mission. I think, again, just to kind of start here with the scope, in the United States today there are somewhere near 1.2 million people infected with HIV and about 40,000 new cases a year. In Africa, there are 26 million people with HIV and 1.4 million new cases a year. So more than two-thirds of the infected population of the world is in Africa, most of it in the dark-colored countries here in the south. As with the rest of the world, there are now very effective anti-retroviral therapies, great drugs. If you take these drugs like you're supposed to, you will not advance to AIDS, you will not pass the infection to a sexual partner. If you're a mom, you will not pass the infection to a baby during birth or through breast-feeding.

The key, though, to anti-retroviral therapy is getting it to the patients, and so there's a number of elements in Africa to scale up and expand the use of these drugs to try to stop the epidemic. Some of that is driven around rapidly diagnosing the patients and identifying them. And then I think a big shift in the last few years is, rather than wait until they start to decline, is to get them immediately on therapy. And then a key element is, once on therapy, they need to monitor it. And that's where viral load comes into play.

The viral load assay that we're developing, just like the viral load assay from the other players, is the way that they monitor that the drugs work. If the viral load stays low, then you know the patient's taking their drugs and you know that the drugs are working well. And so that's really been, kind of, captured here by the WHO. There's a shift away from CD4 testing to viral load. And so along with the effort to grow the use of the drug, they're scaling up viral load testing.

The challenge is, in a country like Africa, and most of them, the access to viral load is very, very limited. Almost all of it is only in a limited number of large central labs. It's difficult for patients to get to locations where they can have their sample taken. It's very difficult to get the samples in time.

Technically, you gotta get the sample within six hours of collection to the lab to run a viral load test under the specs. So that is limited greatly. The turnaround times. We talk about cutting them from a day down to a couple hours. Turnaround for some of the places that are doing viral load is measured in months. It's oftentimes two to three months before they get their answer back. And what happens then is, they can't find the patient.

So somewhere on the order of 30% of more of the results done in the central lab never get to anybody and they lose that patient. So the goal here, then, is to move the testing out closer to the patient. You don't lose them. You get to make a decision immediately. You can adjust therapy. And overall, the program should advance nicely.

The real challenge, though, is trying to do viral load testing in this environment and in the point of care mode. One, right away, it's a quantitative PCR test and it really needs plasma. There are other people who have tried, to date, using whole blood. The Alere cube, which I'm sure you're familiar with, is a whole blood assay. And while it works very nice for diagnosis, the data so far suggests that it doesn't correlate well to plasma. And so the key here is you need to be able to get a plasma sample in a point-of-care mode for this to work very well.

You add that to the fact that the sites we're going to are going to be typically smaller hospitals or larger clinics. They don't have air conditioning, they don't have medical staff. So it needs to be extremely simple. We have to put a bunch of protections in to make sure they follow the procedure carefully. We need to be able to develop a finger stick collection device that will yield plasma.

And then the environment is incredibly challenging. The spec from Gates for this product is that it needs to operate at 40 degrees C with room temperature storage of all the disposables at 40 degrees C. To put that in perspective, med techs in the United States won't work in labs that are 40°, let alone worry about the systems or the reagents surviving it. And so that's one of the really tough challenges here in the ruggedness.

And then we need to be able to develop all that with a very, very low cost to support the mission there, where funds are very, very limited. And then ultimately, the other key element of the spec is they didn't want to deploy a system out to do viral load that couldn't do other tests. And so the ability to do other different assays was really an important element.

So Gates captured all of this into a set of very, very tough specifications which guided the development of the system. And so what has kind of evolved out of that is what we call the Savanna system. It includes the analyzer that you can see sitting on the table over in the front and a disposable cartridge.

To quickly highlight kind of the key elements here, if you look at the overall abilities here, it's a true sampled result system. And it's random access, a two-bay system.

One of the real nice things, though, is that we've built this so that it can handle not only PCR and quantitative PCR, but it will also hand our isothermal HDA assays as well. So we have the option to select and pick the technology that suits the situation the best.

We can do, also, with the system, we can do small panels, multi-target panels. It'll handle a wide range of sample types. If we can get it into a liquid format and get it into the cartridge, it'll run.

Room temperature is probably understating it, given the specs for Africa, but it'll obviously handle room temperature in the developed world conditions. Very, very simple from a maintenance and calibration standpoint. Ruggedness is just a key. And then a long time between failure is critical when you're deploying this, obviously, in settings a long way from any kind of service. Low total cost.

We think also, just based on the simplicity of the design and the built-in safeguards, that it will be CLIA waivable, obviously depending on the assay. If the assay prep requires centrifugation, for example, it wouldn't be. But otherwise it should be CLIA waivable. And then we see all the other assays we'd be developing, we'd be able to hit a turnaround time of 45 to 60 minutes. We're going to shoot for better than that, but right now that looks pretty sure.

So, looking more closely at the actual system, so I'll just walk you through some of the features. You can see what it looks like sitting on the table. We have decided to build the controller into a touch screen, integrated on top of the system that will have very simple graphics to guide the procedure, very simple results display.

The nice thing is this way you don't have to have a separate computer to run it like some systems do, or a telephone to which you have to go through. It's all built in on the system. The other nice thing is, since we've stacked the two bays vertically, it puts the touch screen up at a very nice height from an operating standpoint. So when it's sitting on a bench, it's very, very convenient and easy to use. There's two bays that you can see here, one and two. These operate completely independently.

The other nice thing is we've built in certain safeguards so that while an assay is running, the door will lock and the color will change on the light, so that it's easy to know which bay to go open up if you're going to run a subsequent sample. You can run different assays at the same time, you can start samples at different times. They're effectively completely separate analyzers if you look at it that way.

We've got a built-in bar code reader on the front. You can do all the usual identifications, the cartridge itself, the sample ID, operator ID, depending on what is available in the given setting.

We've built connectivity into it, so it's all of the usual internet connectivity that gets data into LIS's and the like, but it also will include a wireless connection mounted to the back so that we can -- in a setting like Africa where they don't have internet connections, we can transmit the appropriate data and monitor systems from afar.

It also means that we can participate then in the Virena program. So we'll be fully compatible with sending the results here from Savanna up into the Virena cloud. You'll see that there is a bit -- the unit's relatively large, and part of that is that we have a dust-proof housing with no fans, right? And so we need to be able to dissipate the heat without the use of fans. And that's driven around the history of other analyzers that have gone into this segment.

When Cepheid had launched the TB test in Africa, they had a 20 to 25% failure rate on modules due almost entirely to dust, right? There's nothing wrong with the analyzer, per se, but in that environment, dust was getting in the optics. So we have the advantage here, obviously, of designing with that knowledge. And so we built a housing around it to help prevent that from happening.

And then the last thing I'll point here is, while it comes across -- and I would suggest that the reaction we always get is that it's big. When you actually look at the footprint, it's really pretty small. And we'll talk about that a little more later. This unit itself will actually -- the final units will be shorter. So the width will be about the same. They'll take up less depth on the bench.

And then, clearly, you can connect multiple units together. So where there is adequate volume to support it, then you can daisy chain units together. When you do that, they operate -- you can operate them out of a main unit with the other units providing data and feeding to that particular unit's touch screen.

So the cartridge itself, again, looks large at first glance because it's very flat, all right. It's a single piece of plastic molded with a number of chambers and channels over which there's a film that we seal it with. And then we have these -- on the back side you can see these foil blisters. And so what we have, then, is there are some reagents that are in tablet form inside some of the chambers. And then all of the needed liquids for the system are contained in these foil pouches, so that everything that you need to run the assay is essentially contained in the cartridge.

And, again, this is very stable. We have to hit a 40° C storage requirement with no cold chain as part of the requirement for Africa.

So, operating is, again, very, very simple. The sample goes into this well right here, does not require precise pipetting. You just fill it up -- there will be a visual indicator, and then anything above and beyond that will flow into an overflow chamber. And then this cap is placed over the top. The cartridge goes in the analyzer, the door gets shut, and then, depending on the assay, the turnaround time will be between 45 to 60 minutes. You could then run another test of the same type or a different type in the other bay at the same time.

Clearly, if you are using the bar code, then you would also hit the bar code reader on the way through.

So inside, right, there's a number of things happening. And most of it's very simple and designed to be very, very durable. And so what we have, then, are these mechanical cams. And what happens is they go ahead and they squeeze those foil blisters to dispense the liquid reagents up into the cartridge, and then inside we also have, then, the ability to do sonication and provide magnetic forces, including a magnet that moves. So we capture the nucleic acids onto magnetic particles, the magnet moves those particles through the various chambers, where they go through the various washes and the like. Ultimately, these particles are taken into the amplification chamber where we do the amplification and ultimately do the reading.

And so the optics on the system does have five channels, and that's where we get the ability to do the small panels that I mentioned earlier.

So, when we look, one of the things we did here is we laid out the capabilities and also the limitations. And when we went to get feedback from customers, we didn't tell them what assays we thought we should do. We showed them what the system could do and asked them what we should do with it. A lot of what I'll talk about going forward is reflecting that feedback. But, again, the capabilities that you see kind of going across the top, I think, are the -- if you want to call them the obvious ones, or the basic ones. And these here, the low cost, the simple operation and the turnaround time, that enables us to de-centralize, right? Those are key elements to being able to take a system and moving it away from a central lab. But beyond that, we have some other more unique capabilities, I think, and that's down here, as you see below, the ability here to run either of two different technologies for amplification.

And so that gives us the ability to select the technology that works best for that assay. It also means that, essentially, any of assays that we've already developed, either in Lyra or for HDA on Solana or AmpliVue, can be adapted and built into the system pretty quickly. And then, obviously, having the ability to do quantitation extends our capabilities here to a wider range of different assays. And that also couples with the ability here to do multiplex panels.

I don't know if the right word is mini-plex or opti-plex, but three to four targets plus always an internal control. And we are looking, although it's in the early stages, but there are some potential technologies out there that we're assessing that would give us some ability to expand a little bit more broadly and have even wider panels than the three to four that I'm talking about.

And then I mentioned we are developing for the point of care in Africa the ability to take a finger stick sample and convert that to plasma, and that may have utility for other tests in a point-of-care mode at some point.

So, limitations, here again, I think are pretty straightforward, but I think it's important. Obviously, it's a two-bay system with 45- to 60-minute turnaround time. So the throughput for any given unit is somewhere between 16 and 20 tests per shift. That's what you'd have. If you had more than that, obviously you can scale up, but it's not an 80-bay high-volume system.

We have multiplex capability with the limitation I just mentioned. I think there's also some sense that, you know, the overall packaging and the cartridge will take up a fair amount of space. That's clearly offset, though, by the fact that no refrigeration is required at all. And then the other thing we see -- and I've listed here as a limitation with some offset. Anytime we've shown this -- is it failing here? Is that better now?

Anytime we've shown the system anywhere, what we normally get is this initial reaction, right? It's too big for point of care.

But as we really look here, when you look at a bench space and you look at, this is a two-bay system, it actually takes up less bench space than some of the one-bay systems that are out there today. I think that's the real key thing. We've kind of done the skyscraper approach. We figure the bench space is the valuable property, and going up made the most sense in order to stack the bays. And we also don't require any space dedicated to an external PC or other type of controller.

The last limitation here, and what I'm labeling true point of care, the turnaround time we have for the system, as mentioned here, in a waived physician office running flu tests, I think that time is too long. I think we have other solutions that meet the requirement of being able to do it quickly. You know, the ten-minute office visit element of those kind of environments.

And so we don't envision right now that we would be placing this system to do flu RSV, strep testing, in a waived physician office.

All right. So, a little bit of data here in terms of what we're looking at. And I think it's key to understand here this is with very early prototypes, modular systems, not a final version. But it was important for us to assess some of the capability here, to make sure we're going to meet some of the rigorous requirements. And so one of the real challenges of point of care viral load is the sample size is really small. If you look at the Abbot system or the Roche system for viral load, they use a full ML of plasma in order to get to the levels. You can't get that quantity with a finger stick. We're going to have a 50-microliter finger stick plasma volume. And so we needed to be able to meet -- whoops, hit the wrong button -- to meet the World Health cutoff of a thousand. And so we assessed here running a variety of samples at different levels. And what you can see here is that we are able to get well under the thousand. So with room for precision and everything else, we'll be able to meet the goal at the finger stick level.

When you actually go to the quantity that you would use if you were doing venipunctures -- so there will be a second alternative on the viral load, different cartridge that uses a higher volume. You can see in that setting, at 180 microliters, we can get down to 125 copies per ML, which actually is better than the comparable Abbot assay, at even a slightly higher volume. A good indication of the underlying capability of system to perform at the level of sensitivity required for a pretty challenging assay.

One of the other things we did is just look at the overall correlation. So you can see here in this case, we ran a number of samples against the Roche assay. And you can see that the correlation is actually very, very good. And as our scientists pointed out, the actual correlation between Savanna and the Roche assay was very comparable, to Roche to Roche, two versions of the test that they have. So just very good preliminary data that makes us feel good about the potential to move forward with the assay and do the things we think we can do.

The last piece here is, we wanted to test-fire the ability to port assays over, because we do have a nice collection of assays already out there that are molecular.

And so we picked one here that was a tri-plex, so we used the Lyra HSV1 and 2 and VZV. And again, you can see here a variety of very parallel curves demonstrating that the correlation here between what we see with Lyra, you know, and what we see with prototypes of the Savanna assay look very good.

And this was with virtually no optimization. We didn't spend a lot of time trying to make changes. We just took the reagents and changed the form factor that we need to have to individualize them into cartridges and then ran them. And the data looks very promising.

Likewise, Northwestern, through a separate grant right now, is working on a hepatitis C assay. And we asked them to do similar work. We took the reagents that they're working on, and we put them into the Savanna format. And, again, we got really, really nice comparative data that, again, makes us just feel comfortable that when we get the analyzer finalized, the ability to rapidly develop assays looks very, very promising.

So, let me shift gears and kind of just give you the launch timeline here, with all of the usual caveats here. But right now, we believe that at the end of next year, so Q4 '17, we'll complete the viral load assay for the African market. And right about the same time, we'll be getting CE mark for the instrument and the initial assays on it; followed pretty quickly, then, in early '18 with getting FDA clearance on, again, the system and the initial assay.

And then we've made an effort here, again, based on what I showed you and the ability to port assays and everything else, we're going to have a variety of people working on multiple assays at the same time, so that we're able to get -- what we're targeting here is 20 targets by the end of the year. And so if an assay had, you know, HSV 1 and 2 and VZV, that's three targets that we feel we'll have cleared in both Europe and the FDA, 20 at the end of the year. And that will include porting the assays that make sense from our other formats into the Savanna format.

So, what we're going to do in Africa and other parts of the developing world is obviously, Quidel is not set up with a large organization in Africa at this point. We will engage, essentially, a full-service distributor and commercialization partner to help us do that. And we have discussions underway with a number of parties right now. The reason you need the organization is that the target here is going to be really to de-centralize away from the big central labs, and that means going into the smaller hospitals and larger clinics.

You can see, the reason for that is, the patients today that are going to the large facilities can get pretty good service. Right? The people who are farther away, can't. But there is a larger percent -- more than 50% of the patients are actually in those settings. And so the goal is to move out and have the maximal impact in those settings through decentralization.

We'll launch initially with the simpler version, which is the venipuncture, get it established and follow up with the subsequent launch of the finger stick version. We're also looking at potentially TB and other assays that make sense in the same setting here to enhance the value. There's an enormous overlap between TB and HIV in most of these countries.

In the developed world here, I'm going to come back and just kind of reiterate, very recently have gathered a lot of feedback. And, again, I got that feedback by being very open-ended. We just showed the system to a variety of thought leaders. We showed it to our scientific advisory board. I interviewed something on the order of 20 or 25 of our field people to get their assessments. And in all cases, we just showed the kind of capabilities and limitations I just walked you through, and then asked them to tell us what they thought made sense. And, you know, I think I had a narrow view of what we thought we could do. And I think these conversations have really been helpful in seeing what I think is a real nice opportunity.

And it's built around, kind of this environment here. What we see, right, is obviously molecular is de-centralizing very, very quickly, not only the existing players that are there, but a bunch of people that have products in the pipeline; driven, clearly, by clinicians who want the increased accuracy associated with molecular. And to a great degree, the availability of having things 24 hours a day, 7 days a week, as opposed to the classic big, central, molecular lab that ran only during the week. Right? So there's two elements of that.

The other piece that I think you're all aware of is the introduction and growth of syndromic and what I call other multi-assay panels beyond syndromic, multi-syndromic, or whatever you might want to call it, which come into play strongly. But when you look -- and this is the feedback that we got very, very strongly from the people we talked to -- even in the face of all that, there are still some real issues here with -- oops. I keep hitting the wrong button -- with the current systems and the challenges. Right?

When you look at what I'm just calling "full service," right, a system that can do a full range of molecular tests and capabilities. Most of them today still have very high capital costs, when you look at just the equipment up front and the investment.

The flip side are some of what I call true point-of-care systems. And I think those systems are maybe not quite as expensive, but they start to trade off some of the capabilities that are out there. And then associated with, I think, the panel, is there is a lot of these nice panels out here, and they keep focusing on getting bigger and bigger, right? And then the cost per test looks low because you have a $100 cartridge with 20 tests on it, and it looks low, right? But the overall cost per patient is very high, and I think there's a sense from a lot of people we've talked to that there's a diminishing return as you expand that panel on the clinical utility of some of the targets that are on there.

And I think we hear over and over again, it's pretty difficult. If a patient isn't really sick enough to be considered for admission, why are you running a $100 panel of molecular tests, you know, as part of the decision-making process?

We get a lot of this feedback, by the way, through the customers who have our viral culture product. And one of the things that we've heard from them, that a lot of the movement to some of these panel products isn't driven as much by the size of the panel; it's being driven around this 24/7 capability. It's harder and harder to staff viral load -- I'm blowing up here -- the viral load -- or, excuse me, clinical virology department with enough techs to run it. And so there's a big demand to kind of move into this round-the-clock capability here.

There's no doubt, though, I think -- and I think nobody would argue that for the right patient, in the right setting, some of these panels make complete sense. And so as we look and envision where we go, we don't necessarily replace such systems. I think there is a coexistence mentality here of running the right tests for the right patient.

All right. So I'm just going to highlight here, you know, again, based on feedback, two key areas, right, is where we want to target, you know, based on what we've heard from the customers and what we've analyzed. You know, who do we go after, what the menu needs to be associated with those customers. But, obviously, Quidel has a pretty broad set of products here in term of what we have for Sofia, what we have for Sofia 2 and other molecular. And so we will, obviously, co-market these and provide options to the customers. I am going to highlight in a little more detail Solana, just to clarify for everybody where we see the differences between those two systems.

So, again, I know there is a lot of ways -- and you guys have probably spent more time thinking about that than even we have -- there's lots of ways to segment the market. Some people segment it by technology. Is it an immunoassay, or is it molecular? Or they do it by disease state. They do it by infectious disease versus others.

To me -- and I've got a long history of working in point of care -- when you're talking about de-centralized testing and point-of-care testing, I find the best way to segment the market is by customer site, and in particular, the sites where patients present. Right?

And so we've looked at this environment. And when you de-centralize and you look at Savanna's capabilities and limitations, what we really see is a number of de-centralized segments that make the most sense. And our logic on that, right, is that these tests are out of and away from the main lab. They need pretty quick turnaround time, but in most cases, they can wait 60 minutes.

So in a mod complex setting, where they really have a true lab setup, that's more likely to happen than in a waived physician office where there's a patient waiting in a room for ten minutes that can't be held up.

And likewise, there's small hospital labs where we just fit in, from a pure volume standpoint. The interesting thing that we heard in the conversations with the customer is, I was focusing very heavily here based on what I thought we had. And in the conversations I've kind of expanded it out this direction. And in particular what we're hearing is for a subset of what might even be in a larger hospital. If we put together the right menu, that even in the larger hospitals there is a stat lab and rapid response labs and other stations where this appears to be a very good fit, based on the feedback that we're getting. So, again, that's where we want to target.

When we look at the menu here, I'm just going to break it down into two main areas, what I'm just calling "unique assays" for lack of a better term, and then bread and butter. And let me just go through them one at a time here. By "unique" what I mean are those assays which, in effect, get the system placed, right, that get the interest of the customer; we meet some currently unmet need, or we do a better job than the system they have. And we then drive placements kind of built around the kind of more unique capabilities that we have.

Based on feedback right now, what we're hearing very consistently is that what they like is what I'm just, again, probably needs a brand name of some sort at some point, but just what I'm calling the right panels at the right price in the right place. Right?

So these are de-centralized panels with the top targets -- not 20 targets, but the top ones for clinical decision-making that can be delivered very quickly at a very good price, right, and that cover most of what they need, with the idea that they can always -- and several people suggested this -- you can always have a secondary cartridge to expand one layer deeper.

And in fact, those two cartridges together will still be less expensive, okay, than the alternatives that are out there. And likewise, we do envision that it would coexist with other systems that can do broader panels at higher prices. And so just some examples of what we're hearing and envisioning here right now are a panel for STDs. So this is CTNG, trich, and potentially mycoplasma genitalia, all in the same cartridge in 60 minutes, at a competitive price, that kind of product. Vaginosis, very, very similar, a three-part panel. Pharyngitis, that would be similar to our strep complete. So, that would be group A, C and G strep; all, again, run on a cartridge in very short order. And then, of course things like our HSV 1 and 2, VZV combination makes sense.

The other area that we're looking at, again, is the ability to have a de-centralized quantitative system for things like transplant assays and the like, which today all are done in pretty centralized settings no matter where the patient is. And then there's the whole area of high-burden disease that is tied very closely to what we're doing with viral load, but could be expanded to include a variety of other similar infectious diseases that occur in low- and middle-income countries, that are very challenging.

Bread and butter is pretty simple, right? One of the things that I think is really critical when you de-centralize is, unlike a big central molecular lab that can have 25 different systems to run their tests, as you de-centralize, people don't want to have three or four platforms to run their menu. One of the key things is looking at the basic assays that you would add to the menu, so if they bring it in to do one set of assays, they can replace another system and keep life simple. So that enables the consolidation.

The other thing, in some of the settings that we envision, the overall volume is low of any given assay, and so the way to justify and have enough volume is to make sure the menu is broad enough to give you enough tests to make sense to bring in even a low-cost system like Savanna.

And so, again, that's very much your commonly run tests, that I think are typical of almost any of the molecular menus that you see out there.

All right. So, let's look, again, then. So where does this fit in between the two systems? I'm going to go backwards here, because we've been focused on Savanna, so I'll just again highlight what I've said. Savanna's sampled results. Everything's built in the cartridge, sample in, results out. Multiple technologies that are on board. It's a random access system with only two bays. So what it's good at and what it makes sense for is on-demand testing. There is no value in waiting and holding samples to batch on an analyzer like this, because you're just not using bays, right? And so it's clearly not set for high-volume batching. It can do these panels real effectively. It can do quantitative assays where needed. And I think it's very, very likely that we can get a clear waiver with it.

Solana, on the other side, really is, I think, very synergistic with what I just showed for Savanna here, right? It's a little -- it's got a little more hands-on pipetting and steps, right, and is limited to the isothermal. But it gains greatly in terms of the ability here to do batches at a very, very cost-effective level. So 12 samples in essentially a little over 30 minutes, and that makes it really, really optimal for batch testing.

In the obvious way it makes it good for higher volumes, where lower cost is important. But I think it's really, really important to understand this last bullet point here. Everybody in this room knows about the challenges that seasonality creates for Quidel. Doug just addressed it, you know, in his slides earlier in terms of the volume from year to year and what that means.

But I think what people forget is that it provides the same challenge to customers, right? And having the capability and the ability to scale when a season can go greatly -- and not only is it from year to year, but it's within the year. Most of these flu tests occur in one or two months. And then if you have a system like Savanna, or like Gene Expert, or like the Liat you need a lot of bays to handle your busiest day.

Well, what happens the rest of the year when you don't have it? That equipment sits. So your investment in capital to meet peak testing requirements is -- leaves you with a lot of excess capacity the rest of the time. And I think that would be true, and why we don't just see a simple flu test, necessarily, on Savanna.

You look at Solana and it gives you an incredible capability at a very low cost with a very low footprint to greatly scale, right, in those settings; and for customers to meet that need with a very, very small investment and with the ability to handle a huge amount of volume.

So I think that's where the two, in my view, fit very nicely side by side in a lot of settings -- or we give choices to different settings, depending on the way they operate. And so I think -- I didn't appreciate it as much until all of the recent efforts here and feedback. And I'm also giving you the Savanna view of the world, since I'm the Savanna guy. Maybe the Solana guys would have a slightly different bent. But I really see the value of this, and I think it's been proven with the response to the strep test. And I think it'll be more so when we get to flu.

So, last slide here. We've looked a lot of ways. We have a lot of data, you know, from our experience with the physician office and hospital with a lot of our tests, so we have good data on strep and flu and respiratory in general in a variety of areas. We buy certain databases, GHX, EMMES but also very recently, if you haven't seen it, there was a really good report from Kalorama in March of this year. And this report is the first one that breaks down molecular almost exactly the way I showed you those market segments earlier where we're targeting.

And it so it's very nice to match up the trending that they see and what their expectations are. And so I just grabbed from their data a very high level, what they see as the market potential in just the things that we have kind of on the current menu. I'm not even including here -- there was a huge amount -- there's 1.7 billion, right, on the high-burden disease front that's not even in this segment. But if you look here at respiratory women's health here and just part of the HAI, you can see you're approaching $2 billion in potential and actually, their estimates are that somewhere between 20 and 30% of this will happen in just the next several years, as part of the decentralization.

So, without having to agonize too much or extract from the overall molecular market, you can look at a subset and recognize a lot of nice opportunity for what we're developing for Savanna.

All right. I think that's it. Hand it over to Randy.

RANDY STEWARD: Thank you, Bill. Good afternoon, everybody. Thanks for joining us for lunch. Pretty healthy group here, since all the chocolate chip cookies are still out there. So it looks like Rob and I will finish off with some cookies.

Hopefully you can see the passion that Bill has relating to the Savanna project. I wanted to have him talk to you and update you, so we thought it was important to give you an overview of what 2018 and 2019 look like from our molecular platform strategy.

So now let's go back and review the financials and what we're focused on here in the next kind of 18 to 24 months that's going to help us continue to grow this franchise.

The good thing is our long-term financial objectives are intact. You've seen this before in our previous slides. We continue to drive the business from a financial perspective. You all understand under number 1, the low- to mid-teens growth. You saw it in Q2. You're going to see that as we exit 2016, we're going to have several more assays on Solana. We're going to have a Sofia 2 instrument. We look to launch Lyme and Vitamin D here in a short time period. We hope we have a good structure in place to continue to drive this growth into 2017.

Number 2 here, investment and development commercialization in new products and gross margins. As we mentioned, with these new products, all of them have gross margins of 70% or in excess. So as we get the revenue growth out of our molecular franchise, out of the Lyme and Vitamin D, that automatically will grow our gross margins in anticipation of getting gross margins in excess of 65% over the next couple of years.

Operational efficiency, we've talked about this quite a bit. We've given guidance -- I have a further slide we'll talk about a little bit more. But we believe we have a structure in place from a sales and marketing, research and development, G&A, where we will not need to significantly increase our spend over the next couple of years as we see some good growth in the business.

Maximize cash flow. As you've seen, we currently are not a very capital-intensive business. You'll see cap ex being anywhere from $12 million to $20 million a year depending on what we're doing from an operational perspective. But we have in excess of $30 million of add-back noncash expenses. With that, we see this model being able to generate some very good cash flow over the next several years.

And the fifth one is pursue outside opportunities. I have another slide. We'll kind of look at the history of what we've done, and then potential of what some opportunities are going forward.

These are just some financial overview. You've seen these. It's out on our website. Don't need to talk anything more about it, other than we do sit with cash, a little over 155 million as we exited second quarter. We do anticipate over the back half of the year to see some cash flow, so that number should be probably closer to 160 million as we exit the fourth quarter.

I talked a little bit last week relating to some -- where we see the spend for the rest of the year. In research and development, we have -- we believe we're going to target between $39 million and $41 million for the year. What we're seeing, we continue to invest the Savanna platform. We have Sofia 2 that will be at the end of development here at the end of this year. And then also we have seen increase in clinical trials this year versus last year relating to several of the Solana assays, as well as Vitamin D and Lyme. We did give some direction that we thought in 2017, our R&D spend would be slightly less, probably in the mid-$30 million range.

Sales and marketing. As you know, over the last several years we've invested in our -- on the human resource side. We have, we believe now, a good complement of direct sales force, along with our distribution partners that we believe will help support our growth as we look out over the next 24 months. So we do not see any more significant investment on the sales and marketing side.

Last slide here is kind of an overall summary of what we've done from a capital structure perspective over the last five years or so. In 2010 we did diagnostic hybrids, paid approximately 130 million for 45 million in revenue. This really was a pretty strategic acquisition for us, because what we've done with that asset is we've now made that Athens facility our molecular franchise, manufacturing-wise, as well as you've seen that we've now kind of ramped up our bone health and complement business. So we've added another 12 million of revenue into that platform.

So in that facility now we're probably running a little over 60 million in revenue, and it's contributing in the high teens as far as an EBITDA contribution. So, very good investment for us, and we continue to use that. We'll continue to use that facility as we grow our molecular franchise.

We did raise 60 million with a share purchase in 2011. Also bought BioHelix, which is today where we got the AmpliVue and the Solana platform. So that was, again, a very strategic acquisition for us. AnDiaTec, we've added some additional assay capabilities on the molecular side that hopefully you'll see as we get closer to launching Savanna.

We did do a convertible note offering in December of 2014. We added $172 million of cash on the balance sheet at a very reasonable rate, 3.25%, six years. So we certainly plan on using that cash for M&A activity as we move forward. We did do over the last year, as a result of our stock price being a little depressed, that we did opportunistically buy back about 2.5 million shares of stock when it was down in the below $17 range. And we also, in the first quarter of this year, bought back approximately $4.5 million of bonds. Certainly our priority with the cash is to use it for M&A activity.

So, we did, the end of March, did Immutopics a little over $2 million annualized revenue for us. But there again, we're integrating that into our Athens facility. And once we do that in Q4, we believe that -- even with that size of a revenue number, we'll be adding about a million dollars a year in EBITDA contribution.

So, as you can see for us, it's very important that we do the right M&A activity and we make it strategic and important for our long-term growth expectations. We've said that we want non-seasonal revenue, and we do want EBIT contribution.

The good thing is there's some good opportunities out there for us as we continue to look. As I said, we had $155 million of cash on the balance sheet. Our priority is to use that for M&A activity, and hopefully that here over the next 12 to 18 months you'll see that we're going to use that very wisely to help enhance the growth of our business.

So, with that, that completes the formal presentation. So that gives us about a half hour or something for Q&A. So I'll turn it, I guess, over to the group. Kim here is going to help us with the mic.

>> Great, thanks. Thanks for hosting it. I guess a couple questions for Bill, if I may, on Savanna.

First off, just a couple of housekeeping questions. But, just remind us what the device and the instrument is. Additionally, if we think about, kind of, PCR versus HDA, if we think about both capabilities being built into the same instruments, we're not choosing an HDA instrument versus a PCR instrument, right?

BILL FERENCZY: I'll answer this. Can everybody hear me? There we go. So, second question first, no. They'll simultaneously be on board. You either hold a constant temperature for amplification or you cycle it. In either case, we just use the same Peltiers inside. And then from a standpoint of price, looking at -- in the $10,000 to $12,000 range is again, built off the targeting from the Gates specs.

UNIDENTIFIED PARTICIPANT: Okay, great. And then in terms of finger stick plasma, presumably you're going to have to have some sort of additional mechanism to spin it down or separate it somehow.

BILL FERENCZY: Yes.

UNIDENTIFIED PARTICIPANT: So can you elaborate on that? And secondly, I have a follow-up on the HIV viral load.

BILL FERENCZY: Okay. I'll do the first one. I'll give you kind of where we are with it now. And we did a lot of work. If anybody's ever looked at this, there has been longtime efforts to figure out how to do this properly. And we're at a point now where we have a small device here that would act as the collection device.

So it's got a capillary built into the inside of it, and you collect the whole blood into that device. That entire device, then, goes into a small centrifuge, right, a mini centrifuge or whatever you want to call it. And in effect, we are not using any membranes or filters or anything, we're just spinning it, right? And we spin and then siphon off the plasma.

And then that same device connects directly to the Savanna cartridge and goes in the instrument. So even though there's technically a -- you know, a collection and centrifugation step, you have no transfer of materials, and we stay with proven technology, right, to separate the blood out.

So you've got to collect -- you know, it's not as easy as doing, you know, a diabetes-type test, because you've got to get enough whole blood to get 50 microliters of plasma, so it's closer to 175 microliters collected. So you need a pretty good lancet to do that.

But we've done a lot of work to show that that amount is feasible and preferable to venipuncture, at least in the African setting.

UNIDENTIFIED PARTICIPANT: Got it. Sorry, last question on viral load, and I guess the data you showed was 125 copies versus 150 on Abbot, is the LOD. What's your 99th percentile? Forgive me, It's been a little while --

BILL FERENCZY: I would have to go find that for you, but I think the conference level -- there's two cutoffs. One I probably should have mentioned. There's the WHO thousand copies, right, which is the target for Africa and those settings. If you get to the U.S., right, the definition of virological failure is 200 copies, right? And there are people -- there is controversy about whether you really need to go down to the 40 and 50 copies people get. But the actual definition of and real cutoff right now in the guidelines for NIH is 200. So the key here is to demonstrate ultimately that we can do both. But, again, the key one for Africa would be the thousand -- probably in either setting, because I don't think most of those places are going to go pretty simplistic, when they decentralize, right? Just stay below a thousand and everything is good.

Did I answer your question?

UNIDENTIFIED PARTICIPANT: You did. Thank you.

BILL FERENCZY: Okay.

UNIDENTIFIED PARTICIPANT: Hi, thanks. I had another question on Savanna and the commercialization strategy, and the developing market.

In Africa, you have a really well-established competitor there that's also trying to bring a new instrument to market for the de-centralized locations. Can you just talk about the strategy to displace someone that already has a nice lead in the market?

Obviously, the specs can appreciate some of the advantages there. But what are some of the other strategies you can take, leaning on your funding partners or things like that?

>> Yeah, I'm not sure that -- I was in Africa two weeks ago. I'm not sure I would agree that there is an established point-of-care viral load system out there today. In fact, there is a desperate need to get one that really works. There is a de-centralized TB system, right, in the form of Sofia widely dispersed, but still not in what I would call the settings we're going to go for. They have to build rooms in some of the hospitals they're going to in order to meet the environmental requirements. And so when you're looking at viral load, it's still a central lab play and the scale-up right now is drive blood spots to the central lab, because nobody is really meeting it.

And so I think the combination of -- and what -- you know, I think it goes back to the actual specs that Gates had the foresight, I think, to lay out, which is the combination of simplicity, ruggedness, and cost, right, to get you out far enough. And the cost is on both fronts. You've got to keep the capital investment down, and obviously the cost per test low in order to have, you know, a low ownership cost.

UNIDENTIFIED PARTICIPANT: Great. And then I have one for Doug if he's around, and one for Randy.

>> Sure.

UNIDENTIFIED PARTICIPANT: For the pipeline, the alternate site transactions, you expect nice growth over the next -- hey, Doug. So with the alternate site opportunities, you know, do you expect nice growth over the next couple years, but the retail settings were going slower than you would have originally expected. Could you just talk about some of the hurdles there, and what was able to drive the success with the agreement you announced with the second quarter call?

DOUG BRYANT: Yeah, let me qualify what I meant by slow growth. So the growth of the retail clinic segment is slow. For example, Walgreens still only has 444 sites. CVS now has about 1100. When you look at urgent cares are already at 7,000, community health centers are 7500. Everything else is outpacing the retail clinic segment. These folks are setting up their own boxes and all that. I think their bigger competitors are going to be Walmart and Walmart-like places and grocery chains who have pharmacies in the back. So that's what I meant by growth.

In terms of our own approach, we're already in Walgreens. We're already in CVS, but we haven't gotten them to fully understand the value of Virena yet. And so our approach there is, we'll have them surrounded, and then they'll realize what they don't have.

But, for the most part, those two institutions are actually more focused on doing all this themselves, using other sources of data in order to predict -- or not predict, create a data product that they can use to manage their supply chain, manage vaccinations.

They have different strategies, including one of them partnered with Theranos, right? So they have other ideas on what they're working on, and I don't think our approach is quite yet appreciated in the same way that Walmart appreciates it or that some of the other ones.

And in terms of the large close that we just talked about, the 652 stores -- I think that's the actual number -- the key was they want the data. And they want to differentiate themselves from their competitors in their market, in their state. And so having the ability to pull traffic into their store is a big thing.

And what I now know that I didn't know before is that the flu, cough, and cold aisle is the high-margin aisle in that store. It's not the bread aisle, right? So if they can pull you in and get you down that aisle, they do better. And so that's kind of the driver behind all that. And the ability to standardize and to have the quality control across the entire system, the ability to go online through myvirena.com and see all that is what sold them on our particular offering versus others.

And this month, the month of August, we're now training 18 of their folks, and then they're going to go out and use those 18 to train the remaining 600 and whatever. And they want to go right away. This is -- for us, fascinating, because the other situations that we've been involved in, they all wanted to do pilots. And this particular customer said no, we're not doing a pilot; we're doing it. And -- which is great.

So I think it's going to accelerate. We'll have examples pop up throughout the country where this is successful and where people are competing for foot traffic. I think it's going to go. It'll probably take 12 to 18, maybe 24 months before we can see real takeoff. But as you can tell, we're excited about it.

UNIDENTIFIED PARTICIPANT: Great. Last one, for Randy, you mentioned a 30% EBITDA target. Can you confirm the timeline? Is that next year? You gave a little color around R&D. Just maybe a little bit more around sales and marketing, how you leverage that for next year in some of the new product launches.

RANDY STEWARD: Sure. The 30% EBITDA, we always caveat that with the normalized flu season. So if we don't quite get there next year, we certainly believe in the 2018 time period. So, you know, it's a quarterly kind of review. But certainly if we don't get there next year, we'll get there in 2018, for sure.

We will . . . You know, excuse me, Jack. I'm sorry. Your other question was -- oh, margins, I'm sorry. Our margins, we believe, certainly, should be there at 65% next year. As we see the ramp-up on the molecular side, the Solana assays, additional, and also with the Vitamin D and Lyme, we do believe that we can achieve a 65% margin in the 2017 time period. Okay?

UNIDENTIFIED PARTICIPANT: Thank you.

UNIDENTIFIED PARTICIPANT: One over here. Thanks for all the color on Savanna in particular.

My question focuses on whether or not you pursue CLIA waiver on it. and I believe you said it's CLIA waivable. I'm not sure if you fully decided to go after the waiver. In the event the turnaround time stays at 45 to 60 minutes, it seems like a longer turnaround time for most CLIA-waived settings today. Which assays do you think would be appropriate in that environment? I guess the second part of that is, I guess, do you think there's opportunity to get turnaround time down even further from what you've shown?

SPEAKER (): Let me -- I think you caught the subtlety very nicely there. I agree with you that, you know, that I think I said it in the limitations, the current turnaround time probably precludes, for the most part, what I call that point of care in a waived physician office, because I just don't think they have -- the workflow in those settings, it's not samples; it's patient movement, right? But when you move to some of the other segments I did target, I think there is more potential to tolerate that turnaround time.

And even though those are typically mod complex settings, there is just a lot of value in having CLIA waiver in terms of reducing the regulatory burden on the customer. Right? And so the plan would be to -- for as many assays as possible -- pursue CLIA waiver, even though we wouldn't necessarily target a CLIA-waived environment. We hear very clearly from customers that they would like CLIA-waived products in any of the settings, just because of the new ICQP -- whatever. I never say the acronym right. The risk assessment they have to do on non-waived products is a lot of extra work.

And so anytime there's a waived option available, it makes life easier. I wouldn't want to speculate on trying to go faster. Our goal is to go as fast as possible, but right now we're accepting where we are with turnaround time.

>> If patient throughput is between 10 and 12 minutes, which I can't find a facility where it's greater than that, no amount of work starting at 45 to 60 minutes is going to get you down to under 10 minutes.

>> Yeah. I think almost any of the existing systems out there today, even --

>> They're too slow. They're way too slow.

>> I think that's what we'll talk about with Sofia 2 in the back later.

>> On the other hand, if we are successful at improving the sensitivity of our immunoassay products such that they are greater than 98% sensitive relative to cell culture, and are therefore potentially able to be described as confirmatory, then I can have molecular-level performance in five minutes or less. That's the answer for the physician's office.

Our idea, generally -- I think Bill positioned this very nicely, is our idea with all these is to de-centralize. But de-centralize doesn't mean all the way down to the lowest common element, always.

Again, it depends on what the capabilities of your system are, and what the limitations are, and what the customer tolerances around those limitations are. So, I can see somebody waiting in a pharmacy setting, maybe, you know, in the back of an Albertson's, 30 to 40 minutes for a test. I could see that.

But in a physician's office? It's hard to imagine that the physician can tolerate having you sit in that exam room. Right?

>> And I think we were very -- again, I think I mentioned the way we sought feedback. I went in expecting that the turnaround time would limit us even more than that, and I'm hearing quite the opposite. You know, the support of patients presenting to an ED, for example, where, you know, the times are typically longer anyway. Not necessarily in the ED, but a stat lab, a laboratory stat lab.

And then the third element I would add to what Doug is saying is, it also depends on the disease condition you're testing for. If you're a public health clinic worried about an unreliable patient who may not show up for follow-up treatment who has a potential STD, you know, waiting longer for that setting versus somebody who thinks they might have the flu, I think, would potentially dictate the ability to wait longer for an answer.

And we heard that very clearly. I was in a very, very large hospital that had every kind of capability in their lab that you could imagine. And the guy insisted that the ability to support the ED would have him highly interested in a stat lab STD panel, when he does thousands of STD tests every week in the main lab. It was a very -- it wasn't what I expected. And we've heard it, I think, pretty consistently.

>> 90% of our population lives within five miles of a grocery store pharmacy. Physicians are going to compete with that, is where I was going with that. Sorry, Brian. Jump in. Please.

BRIAN: () So you guys talked about shared gains on your slide and the non-correlation to the revenues and the volatility of the season. Where do you think you are in terms of market share, but also as you've seen those share gains, and as we've seen a market extension into some of these other areas, what is the new normal for you guys in a flu season that's obviously continuing to tick up. So what would a normal flu season now look like for you? And then a follow-up for Randy on M&A in a minute.

>> Yes, 85 to 95, what we had on the slide, is the new normal. And to parse the difference between market growth and shared gain is a little bit difficult for us, given that there aren't really good third-party data. When we do data for our board, in order to support our compensation reviews and all that, we actually have to get -- we have to engage a separate third party to go out and do a survey in order to try to calculate market share. I mean, it's really -- because it's a funny market, right? It's a bunch of tests done in just a couple months. So it's not easy to get the data. And the data we do have access to have big missing pieces.

So the only thing I can tell you for sure is, our market share is higher than it was. And we think we could support, based on our own data, somewhere in the 4 to 6% -- at least this time, this year versus last year, in the 4 to 6% range of market share gain. We could be under-calling it and over-calling the market growth. So, that's my best answer. It's not a good answer, Brian, sorry. But that's the one I've got.

BRIAN: () Randy, we've heard a lot about M&A for over a year, at least. Can you give some color on types of deals you're looking at, timing that you think something could get done, size of deals? Obviously not who they are, but just give us some color about when we should expect, you know, you to deploy some of that capital.

RANDY STEWARD: Well, as you know, M&A deals take a life of their own. I would say we've outlined priorities relating to anything that would -- we have a pretty strong commercial infrastructure. So anything that we could utilize that efficiency I think would be very prudent for us. Internationally, we have a little over $25 million business, anything that I think that would help us expand the growth internationally, I think would also be beneficial. As far as timing, we -- as Doug has said, we have a pipeline. We are talking to people to predict a timeline.

I think it would probably not be prudent, Brian, but I can say that we are actively looking. We have $150 million of cash on the balance sheet, and so certainly we would want to utilize that.

>> Things that fit our channel are very interesting. And also things we think that with our channel we could actually grow it better than the company that has it now. And I'll just use an example. We looked really closely at Magellan, because it fits. And it had about 18 million in revenue. We didn't think we were -- we'd have to overpay. We started to look at how that would fit in our channel. We liked it a lot.

We were just -- in this particular instance, we were too slow to come to the number that would have been necessary to get there, you know. Like most of us, we would do just the straight-up analysis and try to figure out what it's worth. That's here. And then you have to figure out what you actually have to pay to get it done, trying to guess who's in the market who also wants it. We didn't guess that other guy wanted it, we didn't. We were a little slow on that one. But that would be a good example of the type of company that we've been looking at, if that helps. You know, nothing massive. I mean, a couple tiny ones. But the tiny ones are easy to get done, but they're also hard to integrate.

UNIDENTIFIED PARTICIPANT: How large is the Savanna developing world market, and what kind of operating margin should the developing world Savanna generate?

>> There's no question it's really large, but I think it's constrained from a margin standpoint on any number of fronts. Right? Some of it, in some cases, assays like viral load as part of our Gates agreement. There's caps in terms of what we've agreed to do in other areas. I mean, TB is the one that is wildly huge, but naturally capped, right? You know, if you look at so much of it is still done by microscopy, you know, at two tests for $4 or $5.

So I think it's tough, right? And I think finding the right combination of assays and, frankly, the right partner who can work off of a distribution margin that is less than what we'd expect anything we do in the U.S. is key. And that's part of our conversations with potential partners there.

Again, I think I showed you the number that Kalorama () had, it's ginormous, 1.6 billion. And looking at what we've done in some exercises between us and Northwestern, you can quickly get to some huge numbers. Just to put it in perspective, I was in Durbin, South Africa two weeks ago. It's a pretty decent-sized city. There's 600,000 HIV patients in Durbin alone. And so it's just mind-boggling when you look at the number of people not yet covered and not yet done. So the scale-up is potentially gigantic and in the multimillions of tests. But there's just a huge number of factors; funding from the various NGOs, and the country governments, and whether they stay with it, whether they just punt and, you know, do it all in the central lab and claim they're getting tests done is real hard to figure right now. We've stayed pretty conservative on what we think the numbers can be.

UNIDENTIFIED PARTICIPANT: Can you give us the range that just gives us a sense as to how big the business could be for you in some long timeframe that doesn't get you into near-term trouble about projections?

>> You know, I'd love to tell you I have that on the top of my head, but I don't. I've been focused on the other element lately. I think we put some stuff together last year, Andy, for some of the discussions. I'm trying to remember what the projections were. They were pretty darn big. But even on relatively safe penetration numbers, you know, we calculated it, I know, by looking at what percentage of the market would go point of care and what percentage of that we would get. And we kept both of those pretty low. I think it was in -- I mean, it was clearly $20 million, $30 million, $40 million pretty easily before you can blink.

>> Yeah. I mean, three years out it's -- it can be a $50 million plus opportunity for us, with the right distribution partner and how you attack the developed world, as well.

UNIDENTIFIED PARTICIPANT: Thank you.

>> All right. Well, we have both the Sofia 2 in the back and Savanna in the front, if anybody, before they go to their next thing, wants to take a look, chat with us about those, we're happy to do that.

>> Great. Thank you.

>> Yeah, thanks for coming. Appreciate it.

(Applause)